

How to Design, Build, and Use Interactive Electrical Stimulation

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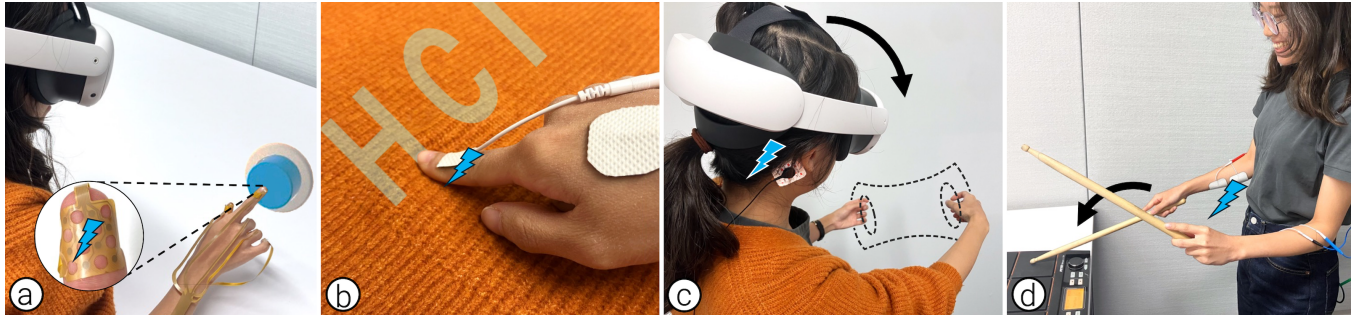


Figure 1: We present a hands-on introduction to the design & engineering of interactive devices that rely on electrical stimulation. Through our course, participants will learn principles and software/hardware pipelines from prior HCI systems that will enable them to create their own: (a) electro-tactile [23], (b) remote electro-tactile [20], (c) balance [1], (d) muscle stimulation [16].

Abstract

Electrical stimulation is now becoming one of the key approaches to creating haptic sensations in interactive experiences. The recent rise of this approach has allowed many HCI researchers to push haptics into more and more domains, including when devices need to be small, portable, or even wearable. At its core, all these techniques share one underlying principle from neuroscience: they act on the users' nervous system to create sensations electrically. As such, using these techniques requires not only getting hands-on experience with hardware, but also learning the fundamental principles, safety, and their possibilities and limitations. In our course, participants will get hands-on experience in using these technologies via hardware toolkits that we will supply.

CCS Concepts

• Human-centered computing → Haptic devices; • Hardware → Emerging interfaces.

Keywords

Electro-tactile, Electrical Muscle Stimulation, Virtual Reality, Mixed Reality, Haptics

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1 Introduction

Electrical stimulation is now becoming one of the key modalities to create haptic feedback in interactive experiences, with more than 300 publications over the last decade in HCI and related fields (e.g., haptics) [3, 7]. The initial surge of electrical stimulation in HCI originated from its ability to create muscle movements, using a technique known as electrical muscle stimulation (EMS) [8, 10, 12, 17, 18]—leading to a course at CHI 2016 on interactive EMS [11].

However, nearly 10 years later, the interactive applications of electrical stimulation expanded well beyond EMS, encompassing a wide range of *electrically-induced sensations*, such as *touch* [20, 25], *body-balance* [1, 15], *visuals* [4], *smell* [2] or even *brain stimulation* [14, 19]—as shown in Figure 1. In fact, of all the interactive domains shown in Figure 1, only the last one has been explored in a CHI workshop before, while all the remainder have not been disseminated in a course-like fashion to the HCI community and have remained as close knowledge inside of labs.

As such, there is an urgent need to revisit the hands-on introduction to electrical stimulation, covering a broader range of perceptual and engineering principles, interactive applications, and, importantly, safety considerations of doing electrical stimulation beyond muscles.

The objective of our course. This course offers a hands-on introduction to this emerging area of haptics, beginning with a short lecture that covers the principles of how electrical stimulation induces sensations (haptics & more), the hardware components' functionality along with their safety features, and a software pipeline. After the lecture, we will have a hands-on component where the participants will use our hardware and software platform to design and implement their own applications of interactive electrical

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stimulation. We will conclude the course with a live demo session where the participants present their own interactive applications.

Broader impact. Our hope for this course is that it also serves to advocate the practicality of interactive electrical stimulation. Participants will learn not only the neuroscience basics but also the design principles to make these devices more practical: (1) mitigating the encumbrance [20, 23, 24]; (2) practical form factors by integrating with existing devices (e.g., smartwatch) [16]; (3) improving the quality of the induced sensation [21]; and even (4) exploring how calibration and other limitations of electrical stimulation can be addressed in the nearby future [5, 22].

2 INTENDED AUDIENCE

This course is intended for researchers, students, and practitioners in HCI or related fields, such as designers, human robotics-interaction practitioners, and anyone interested in haptics and/or AR/VR technologies. It is particularly suited for individuals without prior experience in electrical stimulation or electrical engineering. Participants should have basic Python programming skills, as the course’s toolkit is Python-based; however, no hardware engineering background is required. The course offers comprehensive guidance through hands-on activities and expert-led instruction to ensure participants gain the necessary knowledge and skills.

3 SHORT LECTURE COMPONENT OF OUR COURSE

Our short lecture consists of two parts: (1) principles; and (2) hardware and safety. We cover the foundational knowledge needed to program an interactive control of electrical stimulation. After this short lecture, we proceed to the *hands-on* portion of the course.

3.1 Principle

The lecture begins with the fundamental neuroscience principle of how electrical current passing between electrodes can create a voltage potential within the human body, thereby exciting the nerves [13]. Then we describe how stimulating different regions of the body can result in haptic feedback in a range of modalities by affecting different sensory receptors, discussing recent HCI works that also demonstrate interactive benefits; the examples include electro-tactile feedback [25], remote tactile feedback [20], electrical muscle stimulation [9], galvanic vestibular stimulation [15], and electrical trigeminal stimulation [2]—just to cite a few.

3.2 Hardware and safety measures

We provide each participant with a toolkit, which is a simplified version of hardware from our prior work [20, 22]. We go through the step-by-step hardware mechanism that generates the electrical stimulation.

Safety measures. Importantly, we detail the necessary safety measures implemented on the stimulator [6] and that participants should apply on their future research.

3.3 Lecture materials distributed freely to participants

We plan to distribute the materials from our CHI so that participants can continue their experiments at home and refer to these best-practices; hopefully, speeding up their research ideas and future CHI submissions on this topic.

4 HANDS-ON COMPONENT OF OUR COURSE

After the short lecture, we kick off our hands-on component of the course. In this segment, participants will be assisted with building their own simple interactive experiences based on electrical stimulation—facilitated by our toolkit and its Python API. The goal is that participants will, safely, test their ideas using electrical stimulation of their body. At the end of the course, participant groups can present their application in a round-robin format (2 minutes per group), sharing their creation with the others. We will provide electrical stimulators for the participants to use during the course, along with software libraries and tracking systems, enabling their hands-on prototyping for a wide range of application domains of interactive electrical stimulation described as follows.

Application domains. The participants can design and build tactile feedback interaction with virtual UIs on their fingerpad (Figure 1a). To accelerate their prototyping, they can use self-contained electrode arrays depicted in Figure 1 (a), which we will supply. The participants can alternatively design remote electro-tactile feedback (e.g., stimulating the back of the finger to create the sensation in the fingerpad), as shown in Figure 1 (b). In this way, the participants can explore haptic applications that enable users to simultaneously interact with both virtual UIs and physical materials (Figure 1b). Beyond, the tactile modality, the participants can explore the application of electrical stimulation to modulate users’ sense of balance, i.e., galvanic vestibular stimulation (GVS). For instance, they can implement an interactive GVS that sways the user’s entire body in a driving game (Figure 1c). Finally, the participants are also welcome to implement applications with EMS, where an interactive system directly actuates their body parts to render force feedback, e.g., automated drumming (Figure 1d).

5 TIMETABLE

Our course can work in either two blocks of 75 minutes or one block. Below, we highlight the plan under two blocks of 75 minutes.

Table 1: A timetable of our course.

Content	Duration
Short lecture: principle & safety of electrical stimulation	30 mins
Bio break	5 mins
Hands-on: introduction to a toolkit & initial tests	15 mins
Brainstorming application ideas	10 mins
Hands-on: implementation (with support of instructors)	45 mins
Bio break (get ready for exciting final presentations)	10 mins
Hands-on: participant’s presentations	30 mins
Closing remarks & distributing course materials	5 mins

6 INSTRUCTOR BACKGROUND

Yudai Tanaka is a Ph.D. student at the University of Chicago, advised by Prof. Pedro Lopes. His doctoral research focuses on exploring computer interfaces that output sensory feedback by intercepting the user's nervous system via electrical stimulation. Throughout his research, Yudai has published seven papers related to electrical stimulation at CHI and UIST. Examples include electrical stimulation to create tactile sensations in the palm from the back of the hand [20], and wrist electrical stimulation to produce tactile [22] or force feedback [16], each utilizing a version of the toolchain used in this course. For the past five years, Yudai has been a Teaching Assistant for "CMSC 20300: Introduction to Human-Computer Interaction", supporting the learning experience of over 200 university students. Moreover, as a Teaching Assistant for "CMSC 23220: Inventing and Engineering Interactive Devices", he has assisted students in implementing their own haptic devices, some of which involve electrical stimulation.

Pedro Lopes is an Associate Professor at the University of Chicago. Pedro focuses on integrating computer interfaces with the human body—exploring the interface paradigm that supersedes wearable computing. Over the course of his research, Pedro has published more than 25 papers related to electrical stimulation. Pedro's work has received a number of academic awards, such as seven ACM CHI/UIST Best Papers, Sloan Fellowship and NSF CAREER. His research also captured the interest of the public & media, covered by the New York Times, and was exhibited at Ars Electronica & World Economic Forum. Since joining the University of Chicago, Pedro has been teaching three HCI courses annually. Notably, through "CMSC 23220: Inventing and Engineering Interactive Devices" and "Engineering Interactive Electronics onto Printed Circuit Boards", he has taught over 100 students how to build and program interactive haptic devices and engineer custom printed circuit boards.

Electrical stimulation experience. Both instructors have authored many works on electrical stimulation, including teaching workshops at CHI, IEEE Haptics, and at their university on this topic. Overall, the instructors have extensive experience guiding new learners through the essential knowledge and skills needed to understand and build interactive systems based on electrical stimulation. Both instructors are trained in safety for electrical stimulation.

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